THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS IX — MATHEMATICS · FULL SYLLABUS (PART 1 + PART 2)

Chapters 1–14 | Session 2026-27 | CBSE / RBSE Pattern

Class: 9	Subject: Maths	Set 03	Time: 3 Hours	Max. Marks: 80

General Instructions:

1. This question paper contains 38 questions and is divided into 5 sections — A, B, C, D and E.
2. In Section A, Q. 1–18 are multiple-choice questions and Q. 19–20 are Assertion–Reason based questions of 1 mark each.
3. In Section B, Q. 21–25 are very short answer (VSA) type questions carrying 2 marks each.
4. In Section C, Q. 26–31 are short answer (SA) type questions carrying 3 marks each.
5. In Section D, Q. 32–35 are long answer (LA) type questions carrying 5 marks each.
6. In Section E, Q. 36–38 are case-study based questions carrying 4 marks each with sub-parts of 1, 1 and 2 marks respectively.
7. All questions are compulsory. However, an internal choice is provided in 1 question of Section B, 2 questions of Section C and 2 questions of Section D. An internal choice is provided in all the 2-mark sub-parts of Section E.
8. Draw neat and clean figures wherever required. Take π = 22/7 wherever required, if not stated. Use of calculators is not allowed.
9. Write the serial number of each question before attempting it.

SECTION A — Objective Type — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions (1 mark each). Q19–Q20 are Assertion (A) and Reason (R) questions: select the correct option from (a), (b), (c), (d) given below each.

1. Which of the following points lies in the second quadrant? **1**

(a) (−3, −4) (b) (3, 4) (c) (−3, 4) (d) (3, −4)

2. The distance of the point (−8, 15) from the origin is: **1**

(a) 7 units (b) 13 units (c) 23 units (d) 17 units

3. The graph of the equation x + 2y = 6 cuts the y-axis at the point: **1**

(a) (6, 0) (b) (0, 6) (c) (0, 3) (d) (3, 0)

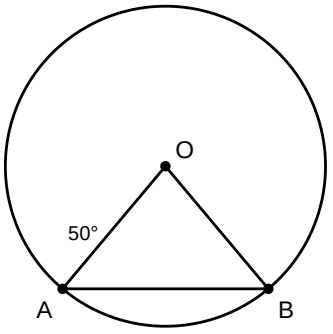
4. Which of the following is a rational number? **1**

(a) √7 (b) √(121/9) (c) π (d) √3 + 1

5. If a + b = 7 and ab = 10, then $a^2 + b^2$ is equal to: **1**

(a) 24 (b) 49 (c) 39 (d) 29

6. In the figure, O is the centre of the circle and ∠OAB = 50°. Then ∠AOB is equal to: **1**



(a) 100° (b) 80° (c) 130° (d) 50°

7. Two chords of a circle which are at the same distance from the centre are: **1**

(a) equal in length (b) parallel

(c) perpendicular to each other (d) diameters

8. The area of a triangle with sides 6 cm, 8 cm and 10 cm, found using Heron's formula, is: **1**

(a) 40 cm² (b) 30 cm² (c) 48 cm² (d) 24 cm²

9. A fair die is thrown once. The probability of getting a multiple of 3 is: **1**

(a) $\frac{2}{3}$ (b) $\frac{1}{6}$ (c) $\frac{1}{3}$ (d) $\frac{1}{2}$

10. If $t_n = 2^n$, then t_6 is: **1**

(a) 12 (b) 32 (c) 64 (d) 36

11. The converse of 'If n is the square of a prime number, then n has exactly 3 factors' is: **1**

(a) If n is a prime number, then n has exactly 3 factors.

(b) If n has exactly 3 factors, then n is the square of a prime number.

(c) If n has exactly 3 factors, then n is a prime number.

(d) If n is not the square of a prime, then n does not have exactly 3 factors.

12. A counterexample to the claim '$4^n + 3$ is prime for every positive integer n' is: **1**

(a) n = 2 (b) n = 4 (c) n = 3 (d) n = 1

13. A student scores 50, 60 and 80 in three components which carry weights in the ratio 1 : 2 : 7. The weighted mean score is: **1**

(a) 70 (b) 75 (c) 73 (d) 63.3

14. The surface area of a sphere of radius 7 cm ($\pi = 22/7$) is: **1**

(a) 308 cm² (b) 154 cm² (c) 616 cm² (d) 1437 cm²

15. To find gcd(m, n) by searching for common divisors, it is enough to check numbers up to: **1**

(a) min(m, n) (b) max(m, n) (c) m + n (d) m × n

16. In a parallelogram ABCD, AB = (2x + 3) cm and CD = (5x − 6) cm. Then the length of AB is: **1**

(a) 15 cm (b) 9 cm (c) 7 cm (d) 6 cm

17. The diagonals of a quadrilateral ABCD are 12 cm and 18 cm. The perimeter of the parallelogram formed by joining the midpoints of its sides is: **1**

(a) 60 cm (b) 15 cm (c) 45 cm (d) 30 cm

18. The sum of the first five terms of the AP 3, 7, 11, … is: **1**

(a) 50 (b) 55 (c) 45 (d) 65

19. **Assertion (A):** The probability of an event can be 1.2. **1**

Reason (R): The probability of any event lies between 0 and 1, both inclusive.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

20. **Assertion (A):** A quadrilateral in which one pair of opposite sides is equal and parallel is a parallelogram. **1**

Reason (R): In such a quadrilateral the diagonals bisect each other.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION B — Very Short Answer (2 marks each) — 5 questions · 10 marks

Internal choice is provided in one question.

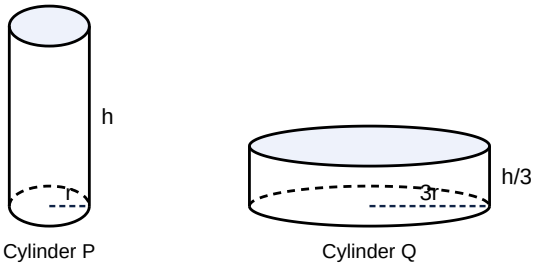
21. An internet café charges ₹20 for the first hour and ₹12 for every additional hour. Write an expression for the charge for h hours (h ≥ 1) and find the charge for 5 hours. **2**

22. Without actual division, decide whether $\frac{13}{60}$ has a terminating decimal expansion. Give a reason. **2**

OR

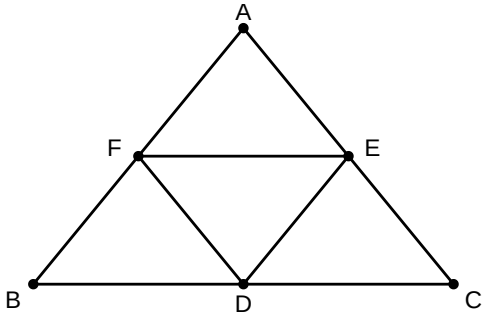
Find two rational numbers between 0.75 and 0.8.

23. The radius of cylinder Q is three times the radius of cylinder P and its height is one-third of the height of P. Find the ratio of (i) their curved surface areas, (ii) their volumes (P : Q). **2**



24. Execute the ‘divisors’ algorithm (check every j from 1 to n and add j to the list if it divides n) for n = 24 and write the final list. **2**

25. In the figure, D, E and F are the midpoints of BC, CA and AB. If AB = 12 cm, BC = 16 cm and CA = 10 cm, find the perimeter of △DEF. **2**



SECTION C — Short Answer (3 marks each) — 6 questions · 18 marks

Internal choice is provided in two questions.

26. Express 3.1666… (6 repeating) in the form p/q in lowest terms. **3**

27. A box contains 3 red, 4 blue and 5 yellow marbles. One marble is drawn at random. Find the probability that it is (i) blue, (ii) not yellow, (iii) red or blue. **3**

OR

A bus ran late on some of the 50 days of a month. The minutes it was late are recorded:

Minutes late	0	1	2	3
Days	18	20	8	4

Find the probability that on a day chosen at random the bus was (i) on time, (ii) at least 2 minutes late, (iii) at most 1 minute late.

28. Draw the graphs of the equations x + 2y = 6 and 2x − y = 2 on the same grid and use them to find the solution of the pair. **3**

29. The monthly expenses of a family are: food ₹6000, rent ₹4500, education ₹3000 and others ₹1500. Find the total expense and the percentage share of each item in a 100% stacked bar. **3**

30. A chord AB of length 16 cm and a chord CD of length 12 cm are drawn in a circle of radius 10 cm. Find their distances from the centre and show that the longer chord is nearer the centre. **3**

OR

Using Heron's formula, find the area of a triangle with sides 13 cm, 20 cm and 21 cm.

31. Prove that the opposite angles of a parallelogram are equal. **3**

SECTION D — Long Answer (5 marks each) — 4 questions · 20 marks

Internal choice is provided in two questions.

32. A shopkeeper draws the plan of a stall on a grid with corners A(−4, 1), B(2, 1), C(0, 5) and D(−2, 5). **5**
(a) Plot the points and join them in order. (2 marks)
(b) Show that AB ∥ DC and that AD = BC. (1½ marks)
(c) Name the quadrilateral and find its area. (1½ marks)

33. How many multiples of 6 lie between 100 and 600? Find their sum. (2½ + 2½) **5**

OR

A traffic survey recorded the vehicles passing a crossing in an hour:

Vehicle	Two-wheeler	Car	Bus	Truck
Number	240	120	60	80

A vehicle is chosen at random. Find the probability that it is (i) a two-wheeler (1), (ii) not a car (2), (iii) a bus or a truck (2).

34. State and prove the Baudhāyana–Pythagoras converse: if $a^2 + b^2 = c^2$ for the sides of a triangle, then the triangle is right-angled. **5**

OR

A goldsmith melts 9 units of gold of purity 12 varṇa, 5 units of purity 10 varṇa and x units of purity 14 varṇa. The mixture has purity 12 varṇa. Find x.

35. The sum of the length and the breadth of a rectangular plot is 17 m and its area is 72 m². Form a quadratic expression in the breadth x, factorise it and find the dimensions. **5**

OR

The area of a rectangular carpet is $(6x^2 + 17x + 12)$ m² and its breadth is $(2x + 3)$ m. Find its length and, for x = 2, its perimeter.

SECTION E — Case-Study Based Questions (4 marks each) — 3 questions · 12 marks

Read each case carefully and answer the questions that follow. Internal choice is provided in every 2-mark sub-part.

36. A mobile shop sold items to 120 customers in a week: mobile phones 45, laptops 30, tablets 25 and accessories 20.

(i) Find the probability that a customer chosen at random bought a laptop. 1

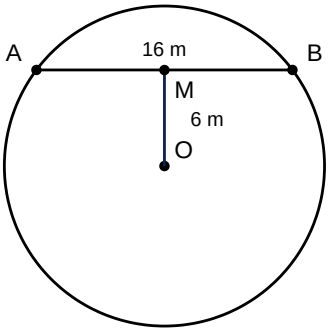
(ii) Find the probability that the customer did not buy an accessory. 1

(iii) Find the probability that the customer bought a mobile phone or a tablet. 2

OR

(iii) Another 30 customers bought laptops. Find the new probability that a customer chosen from all customers bought a laptop.

37. A circular pond has centre O and radius 10 m. A fence AB, 16 m long, is a chord of the pond. M is the midpoint of AB and OM = 6 m.



(i) Find the length AM. 1

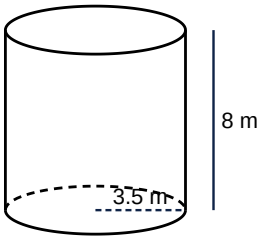
(ii) Verify that the radius of the pond is 10 m. 1

(iii) A second fence CD is placed at a distance of 8 m from O. Find its length. 2

OR

(iii) Find the area of the triangle OAB.

38. A school has an open cylindrical water tank of radius 3.5 m and height 8 m (π = 22/7).



Water tank

(i) Find the curved surface area of the tank. 1

(ii) Find the volume of water the tank can hold. 1

(iii) Express the capacity of the tank in litres (1 m^3 = 1000 litres). 2

OR

(iii) The outer curved surface is painted at ₹40 per m^2. Find the cost.